

Active Fire Sources Classification SOP

Author: Chinmayee P

Link to the application: [Active Fire Sources Classification](#)

Scope: Satellite data extraction, model prediction plotting, manual labeling, and feedback-driven model retraining.

Objective: To standardize the end-to-end workflow for satellite data extraction, model prediction plotting, manual labeling, and feedback-driven model retraining to ensure high-accuracy geospatial intelligence.

Prerequisites

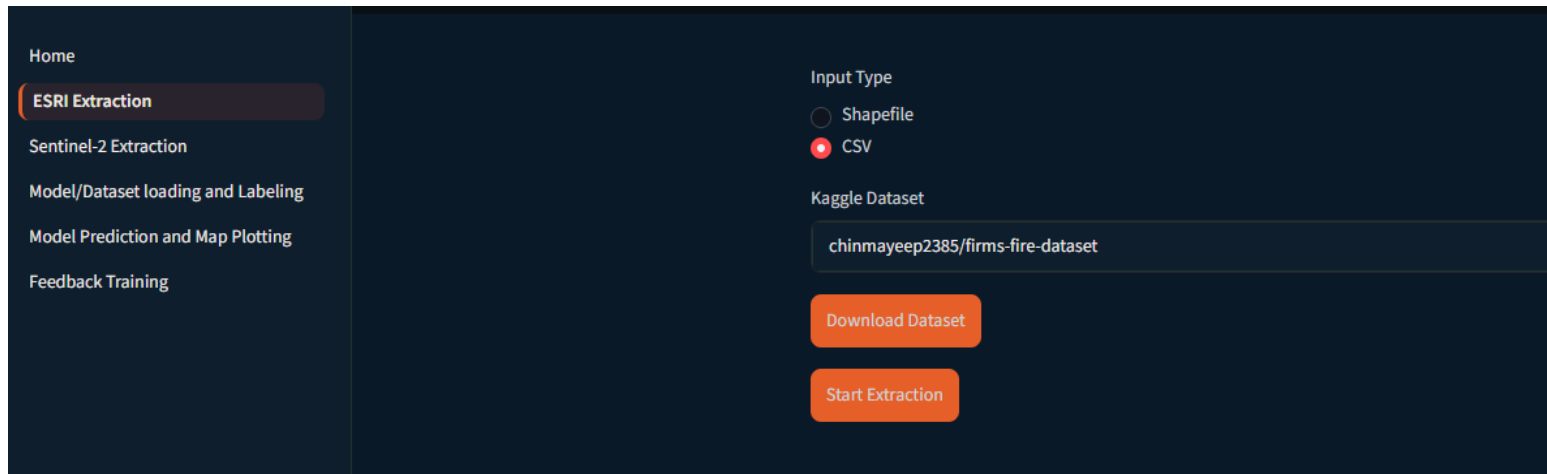
- Active Kaggle account with API access enabled
- Access to the Active Fire Sources Classification HuggingFace Space
- Permissions to create and publish public Kaggle datasets and models

Glossary

Binary Classification	Classifies each satellite point as vegetation or others.
Multiclass Classification	Classifies fire sources into vegetation, mine, factory, solar, and urban.
ESRI Data	Satellite imagery and basemap data from ESRI's geospatial platform.
Sentinel Data	Satellite imagery from the ESA Sentinel satellite constellation.
Feedback Training	Retraining model weights using human-corrected labels.
Model Slug	Unique identifier for a Kaggle model, used in URLs.

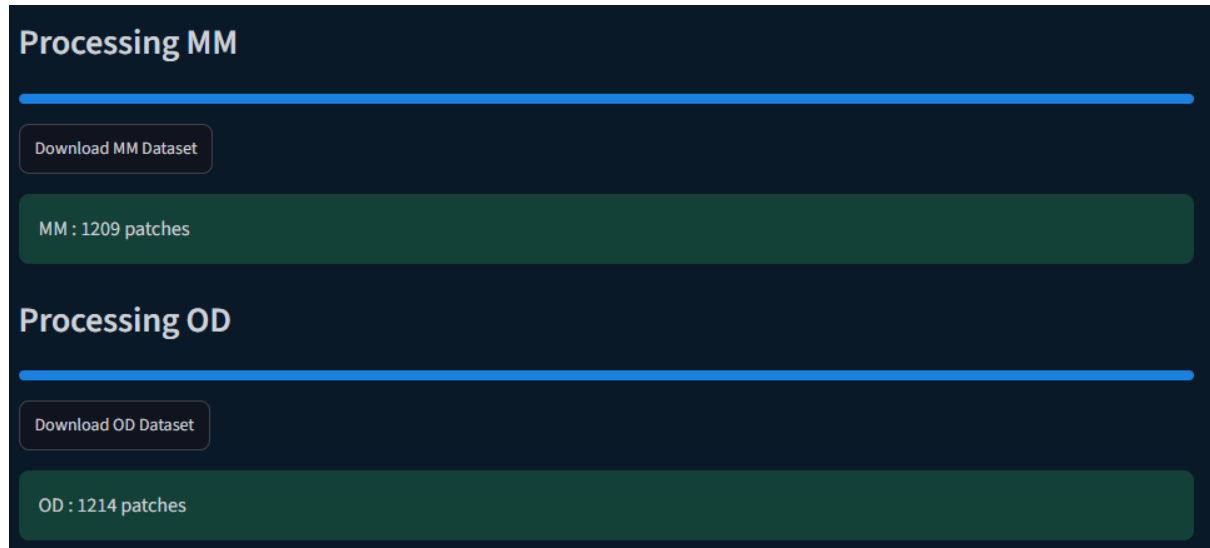
Data extraction

1. Navigate to the “ESRI Extraction” or “Sentinel-2 Extraction” page.
2. Enter a public kaggle dataset URL, in the format “kaggle-username/dataset-slug” which either contains a shape file or csv file with latitude and longitude columns and click on “Download Dataset” and once the dataset is loaded click on “Start Extraction”.



The screenshot shows a web interface for data extraction. On the left is a dark sidebar with navigation links: Home, ESRI Extraction (highlighted with an orange bar), Sentinel-2 Extraction, Model/Dataset loading and Labeling, Model Prediction and Map Plotting, and Feedback Training. The main area has a dark background. Under 'Input Type', there are two radio buttons: 'Shapefile' (unselected) and 'CSV' (selected with a red dot). Below this is a text input field labeled 'Kaggle Dataset' containing the text 'chinmayeep2385/firms-fire-dataset'. At the bottom of the main area are two orange buttons: 'Download Dataset' and 'Start Extraction'.

3. After extraction completes, you can download the extracted dataset as a zip file by clicking on “Download Dataset”.



The screenshot shows two sections of a web interface. The top section is titled 'Processing MM' and features a blue progress bar at the top. Below it is a button labeled 'Download MM Dataset' and a green bar indicating 'MM : 1209 patches'. The bottom section is titled 'Processing OD' and also features a blue progress bar. Below it is a button labeled 'Download OD Dataset' and a green bar indicating 'OD : 1214 patches'.

4. Repeat step 1 and 2 for ESRI and Sentinel-2 data extraction.
5. (Optional step) Upload the downloaded dataset in kaggle, and make the dataset public. It will be easier while loading the dataset in future.

Note: Verify the downloaded zip is not empty.

Expected output

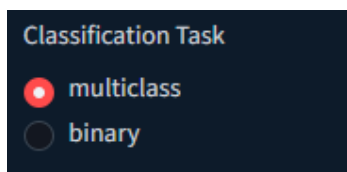
Two downloaded zip files, one for ESRI data, one for Sentinel data

Troubleshooting

Issue	Resolution
Download Dataset button unresponsive	Refresh the page and retry. Check internet connection.
Downloaded zip is empty or corrupted	Re-run extraction. Do not proceed with corrupted data.

Model prediction and map plotting

1. Complete the steps of the [data extraction](#) section for both ESRI and Sentinel data.
2. Navigate to the “Model/Dataset loading and Labeling” page.
3. Select the type of classification, binary or multiclass.

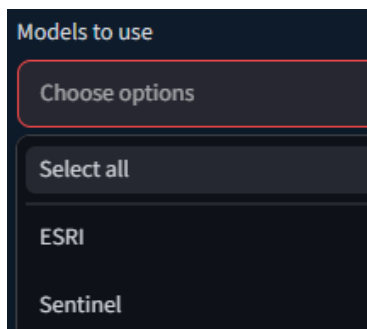


Classification Task

☒ multiclass

☐ binary

4. Select the models to be loaded.



Models to use

Choose options

Select all

ESRI

Sentinel

5. Enter the public kaggle model URLs, in the format “kaggle-username/model-slug/framework/variant-name” and then click on “Load Models”.



ESRI model URL

chinmayeep2385/esri-multi-class/pyTorch/class-weights

Sentinel-2 model URL

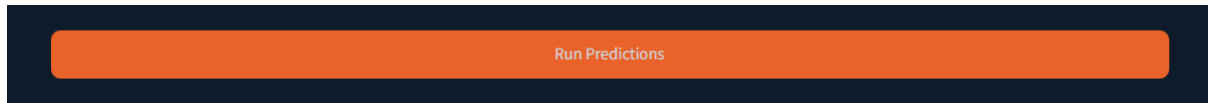
chinmayeep2385/sent2-comb-multiclass/PyTorch/resnet-2stream-4band

Version tag

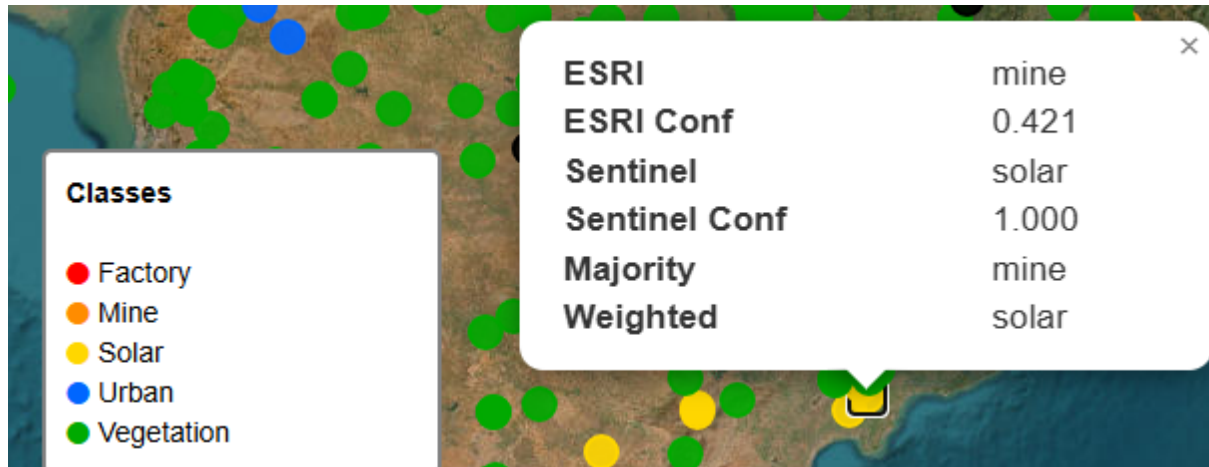
latest

Load Models

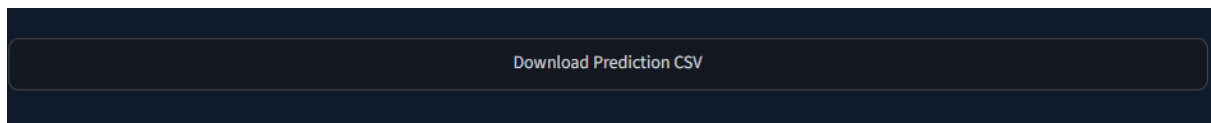
6. Navigate to the “Model Prediction and Map Plotting” page, and click on “Run Predictions”.



7. Prediction statistics and results will be displayed, and will be plotted on ESRI basemap.



8. Model prediction can be downloaded as a csv file by clicking on “Download Prediction CSV”.



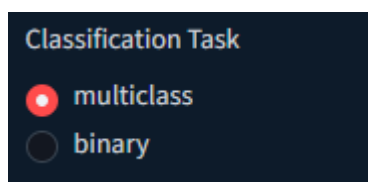
Expected output

- Prediction statistics displayed on-screen (accuracy, class distribution)
- Model predictions plotted as points/polygons on ESRI basemap
- Downloaded predictions CSV with columns: coordinates, predicted class, confidence score

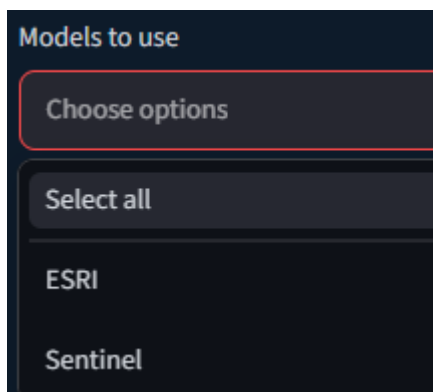
Note: If model predictions appear uniformly wrong (e.g., all classified as others), then verify the classification type matches the loaded model.

Model prediction and manual labeling

1. Navigate to the “Model/Dataset loading and Labeling” page.
2. Select the type of classification, binary or multiclass.



3. Select the models to be loaded.



The screenshot shows a dark-themed interface titled "Models to use". It contains a list of options: "Choose options" (highlighted with a red border), "Select all", "ESRI", and "Sentinel".

4. Enter the public kaggle model URLs, in the format "kaggle-username/model-slug/framework/variant-name" and then click on "Load Models".



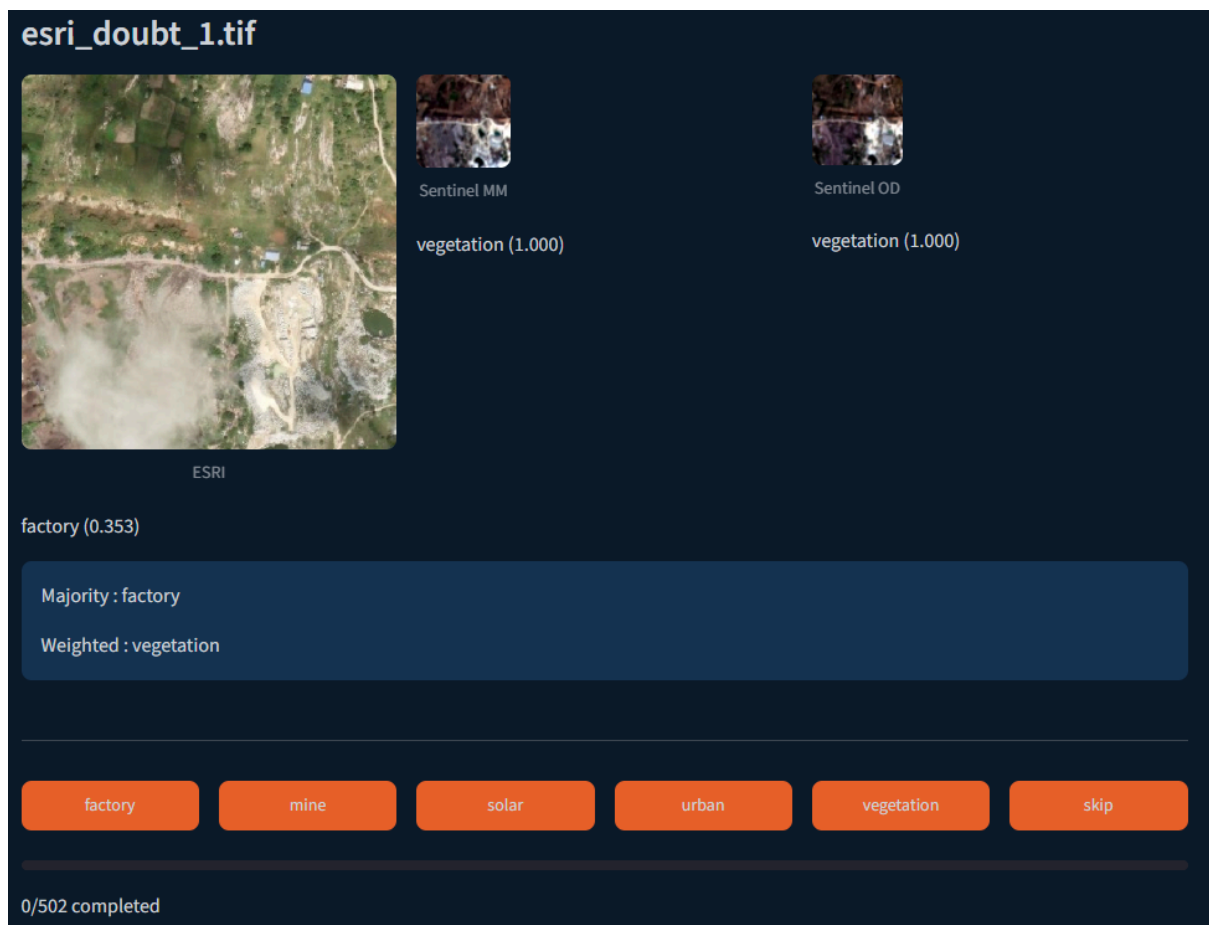
The screenshot shows a dark-themed form for entering model URLs. It has three input fields: "ESRI model URL" with the value "chinmayeep2385/esri-multi-class/pyTorch/class-weights", "Sentinel-2 model URL" with the value "chinmayeep2385/sent2-comb-multiclass/PyTorch/resnet-2stream-4band", and "Version tag" with the value "latest". At the bottom is an orange button labeled "Load Models".

5. Enter the public kaggle dataset URLs, in the format "kaggle-username/dataset-slug" and then click on "Load Datasets".

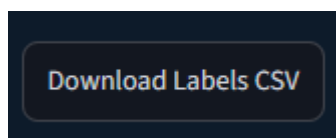


The screenshot shows a dark-themed form for entering dataset URLs. It has three input fields: "ESRI dataset" with the value "chinmayeep2385/esri-doubt-dataset", "MM dataset" with the value "chinmayeep2385/sent2-mm-doubt-dataset", and "OD dataset" with the value "chinmayeep2385/sent2-od-doubt-dataset". At the bottom is an orange button labeled "Load Datasets".

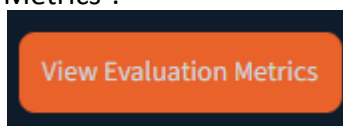
- Based on the classification type chosen and models loaded accordingly, the model predictions and correction buttons will be displayed.



- The feedback labels, including the correct label, is saved as a csv file and can be downloaded by clicking on “Download Labels CSV”.



- Performance metrics of the models can be displayed by clicking on “View Evaluation Metrics”.



Labelling guidance

- Override a prediction when you have clear visual or contextual evidence of a misclassification.
- If unsure, leave the model prediction unchanged and click on “skip”.

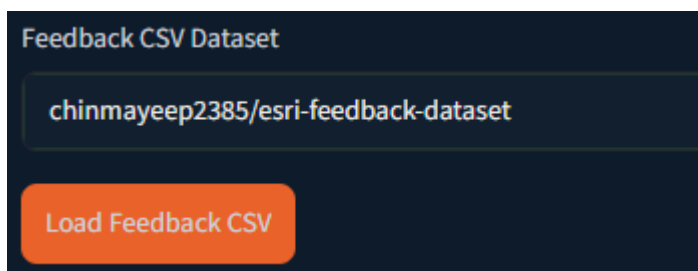
Expected output

- Feedback labels CSV with columns: record ID, model prediction, correct label
- Evaluation metrics (precision, recall, F1-score etc.)

Note: Save the labels CSV immediately after labeling. Closing or refreshing the page may discard unsaved corrections.

Feedback training

1. If you already have a feedback labels csv file, then make sure to complete steps 1 to 6, from [Model prediction and manual labeling](#) section.
2. If you do not have a csv file then complete all the steps from [Model prediction and manual labeling](#) section, and save the csv file as a public dataset in kaggle.
3. Enter the public kaggle url, in the format “kaggle-username/dataset-slug”, of the feedback labels csv dataset, and click on “Load Feedback CSV”. (keep in mind the type of classification model loaded)

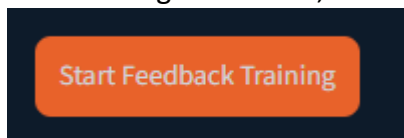


Feedback CSV Dataset

chinmayeep2385/esri-feedback-dataset

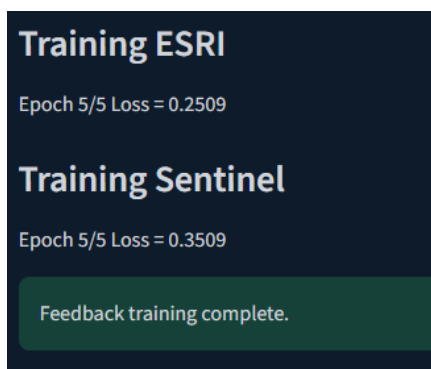
Load Feedback CSV

4. After loading the csv file, click on “Start Feedback Training”.



Start Feedback Training

5. After the feedback training is completed, you can download the updated models (model weights) by clicking on “Download Updated ESRI Model” and “Download Updated Sentinel Model”.



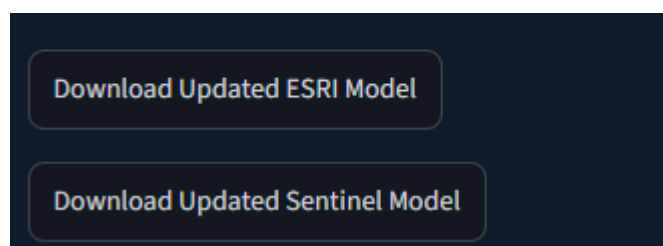
Training ESRI

Epoch 5/5 Loss = 0.2509

Training Sentinel

Epoch 5/5 Loss = 0.3509

Feedback training complete.



Download Updated ESRI Model

Download Updated Sentinel Model

Expected output

- Two updated model weight files, one for ESRI, one for Sentinel (or one file, based on selection of models)
- Updated weights are used automatically the next time predictions are run
- The model file name should be like sentinel_binary_fb322_20260625_065937, type-of-data_type-of-classification_feedback-samples-count_date-yyyy-mm-dd_time.

Note: Always verify the classification type (binary vs. multiclass) matches the models loaded before starting feedback training, otherwise the training will produce incorrect results.

Troubleshooting

Issue	Resolution
Feedback CSV fails to load	Ensure the Kaggle dataset is set to Public and the URL format is correct.
Training does not start	Verify models are loaded and the CSV is not empty.
Accuracy drops after retraining	Check for labeling errors in the CSV. Avoid training on fewer than 200 corrected labels.